Mise en Place

Verified hierarchical bimanual manipulation on Intel OpenVINO

LabLab AI Infra Summit | Intel track | Online submission

Platform: MuJoCo dual SO-101 (12-DoF) dinner-table simulation

Stack: SmolVLM2-256M planning + validation gate + geometric executive + LoRA BC / INT8

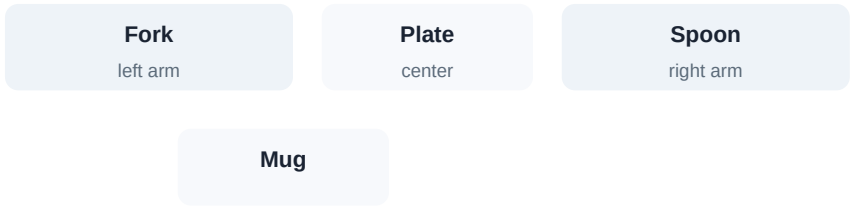


Problem, tasks, and research question

What are we trying to solve and measure?

- Scene: fork, spoon, plate, mug on a dinner table; two SO-101 arms.
- T1: place fork (left arm) and spoon (right arm) on plate in parallel.
- T2: right arm grasps spoon, hands off to left arm, places spoon on left.
- Research Q: Can we combine real multimodal planning with Intel OpenVINO
- deployment while keeping a verifiable geometric safety boundary?

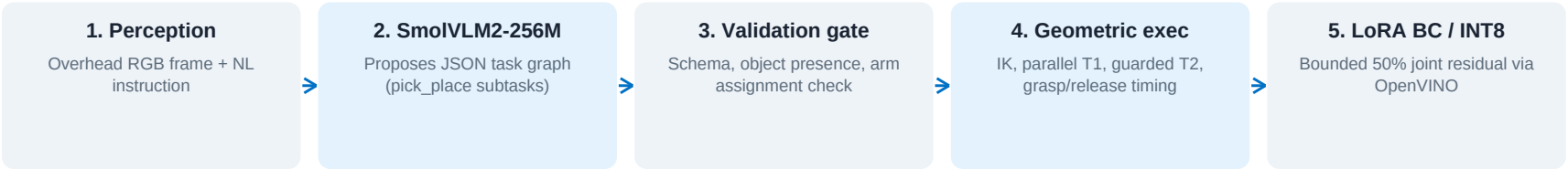
Task layout (top-down)



T2: spoon moves R -> L after handoff

System architecture

Four-layer pipeline from perception to actuation

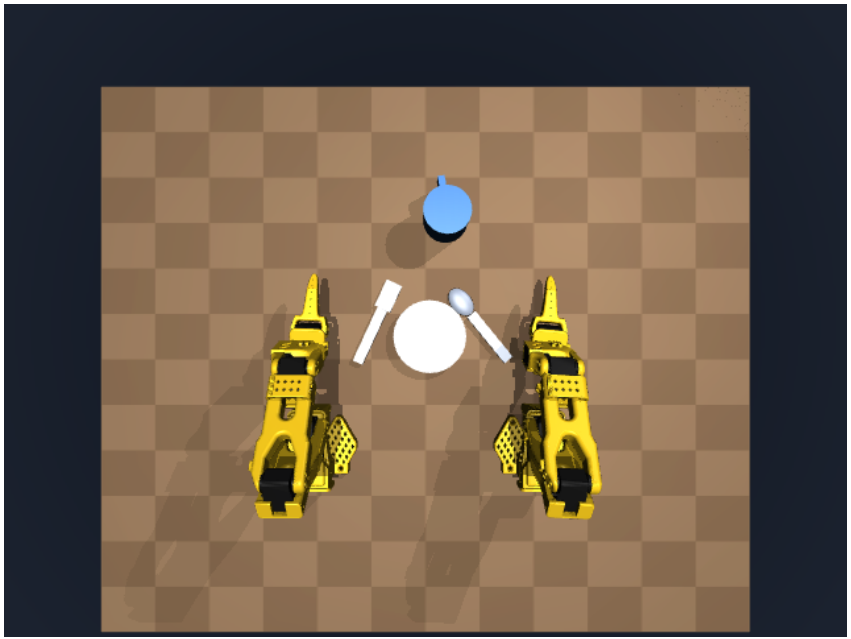


- Invalid VLM plans trigger deterministic rule-based fallback (never sent to motors).
- Gripper open/close commands are never overridden by the learned policy.
- Joint residuals clipped to 0.15 rad max deviation from geometric target.

Simulation and robot setup

MuJoCo environment, actuation, and sensing

- Robot: 2x SO-101, 6 joints + gripper per arm (12-DoF total).
- Control: joint position targets at 50 Hz; weld-based grasp abstraction.
- Randomization: object XY pose, plate position, lighting per episode.
- Observation (42-dim): qpos, qvel, TCP poses, object poses, holding flags, phase.
- Action (12-dim): joint targets for left then right arm.
- Cameras: overhead (planner) + demo view (video recording).



T1 final state (seed 3, overhead view)

Training data and LoRA fine-tuning

Expert demos -> augmented dataset -> 3-stage training



Metric	Value	Detail
Expert success	119/120	1 failed episode excluded
Holdout MAE	0.0105	Joint fine-tune best
Model params	97,132	17% LoRA adapters
HF datasets	0 rows	No compatible 42/12 dims

Experimental results

Stress-tested across randomized seeds with contact audit

T1 parallel
100/100 fork+spoon placed

T2 handoff
50/50 R->L transfer

Collisions
0 70 audited eps

Hybrid INT8
10/10 @ 50% authority

- T1 placement error: fork mean 4.1 mm, spoon mean 3.9 mm (100-seed eval).
- T2 placement error: mean 6.4 mm; max plate displacement 9.2 mm in contact audit.
- Pure BC closed-loop: 0/10 without geometric guard (reported separately).
- One T1 episode used sequential retry fallback; all 100 completed successfully.

OpenVINO deployment and quantization

Export path: PyTorch -> ONNX -> IR -> NNCF INT8



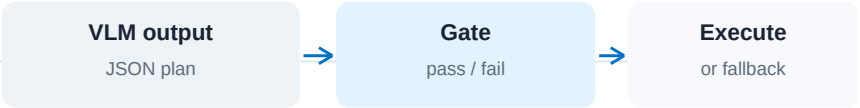
Precision	Size	iGPU p50	Hybrid success
FP32	419 KB	3.13 ms	8/10 @ 50%
FP16	100 KB	2.93 ms	9/10 @ 50%
Static INT8	174 KB	1.19 ms	10/10 @ 50%
Weight INT4	75 KB	5.06 ms	10/10 @ 50%

INT8 iGPU: 2.63x faster than FP32; async throughput 1,189 vs 521 FPS

Multimodal planning and safety gate

SmolVLM2-256M as proposal-only planner

- Input: overhead PNG + text (e.g. 'put fork left, spoon right of plate').
- Output: JSON with scene object flags and pick_place subtask list.
- Gate checks: valid schema, objects present in sim, correct arm assignment.
- Measured episode: valid plan, gate passed, T1 executed without fallback.
- Inference: ~46 s per plan on CPU (256M model).

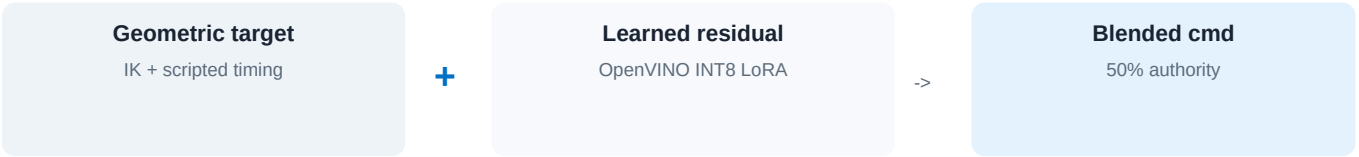


Failed grounding tests (reported honestly):

Structured counterfactual: 0% | Binary yes/no: 50%

Hybrid control mode

How learned inference integrates without breaking safety



- Formula: cmd = geo + blend * (learned - geo), blend=0.5, clip +/- 0.15 rad.
- Gripper joints (indices 5, 11) always copy geometric command exactly.
- Why hybrid: pure BC is 0/10; geometric alone is 10/10; INT8 hybrid is 10/10.
- Finding: static INT8 regularizes destabilizing residuals vs FP32 at high authority.

Limitations, artifacts, and demo

What we ship and what we do not claim

- Limitations: no NPU on host; not end-to-end VLA; VLM not visually grounded.
- Safety: geometric executive is always the boundary; learned model is residual only.
- Video: submission_demo.mp4 (T1 seed 3 + T2 seed 0 + hybrid INT8).
- Artifacts: eval_100seed.csv, contact_audit JSON, inference_matrix.json.
- Scripts: record_demos, train_lora_bc, export_openvino_int8, eval_hybrid_sweep.
- Thank you. Questions welcome.

Mise en Place: verified multimodal manipulation on Intel OpenVINO